

PROVIDENCE HARBOR, RI, USA

WMO: 997278

Lat: 41.807N

Lon: 71.401W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.0	(c) 14.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.6	(k) 32.6	(l) 23.5	(m) 29.2	(n) 11.3	(o) 330	(p) 0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.2	(c) 90.0	(d) N/A	(e) 86.7	(f) N/A	(g) 84.0	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.2	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.5	(b) 20.8	(c) 18.4	(e) DB	(f) 5.8	(g) 94.9	(h) 5.0	(i) 2.0	(j) 2.2	(k) 96.3	(l) -0.7	(m) 97.5	(n) -3.5	(o) 98.6	(p) -7.2	100.0	
			(e) WB	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	30.1	32.3	39.1	50.2	59.5	68.5	75.8	73.6	66.5	56.1	45.4	37.9
	DBStd	17.08	9.64	8.33	8.07	6.63	7.18	6.64	4.69	4.96	6.13	7.05	7.86	8.89
	HDD50	2137	618	498	350	76	5	0	0	0	0	23	181	386
	HDD65	5259	1083	916	805	444	203	38	0	2	54	286	589	839
	CDD50	3245	1	1	10	84	299	556	801	731	495	213	42	12
	CDD65	892	0	0	0	1	33	144	336	268	99	11	0	0
	CDH74	5852	0	0	1	22	219	907	2452	1736	476	39	0	0
	CDH80	1684	0	0	0	9	55	293	768	453	102	4	0	0
(14) Wind	WSAvg	8.5	8.4	8.9	10.0	10.1	9.3	8.8	8.2	7.6	7.2	8.0	7.8	8.2
(15) Precipitation	PrecAvg	45.4	3.9	3.5	4.3	4.0	3.6	3.3	2.9	3.7	3.9	3.8	4.3	4.3
	PrecMax	67.5	11.7	7.2	8.8	12.8	8.4	11.1	8.1	10.1	7.9	15.4	11.0	10.7
	PrecMin	25.4	0.5	0.4	0.6	1.5	0.7	0.2	0.4	0.7	0.9	0.4	0.4	1.0
	PrecStd	7.6	2.2	1.5	2.0	2.0	1.8	2.2	1.6	2.2	1.9	2.8	2.1	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	59.6	67.5	75.4	86.4	93.2	93.3	92.7	87.6	78.2	67.9	64.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	54.7	51.8	59.1	68.3	79.5	87.1	90.4	87.8	82.7	73.7	63.9	59.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	48.8	47.4	54.4	64.3	75.2	82.9	87.2	84.7	79.1	70.0	60.4	55.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	44.0	43.6	50.5	60.7	71.2	78.9	84.4	82.2	76.2	67.2	57.8	51.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.6	14.0	14.9	15.1	14.5	14.2	13.9	14.4	13.8	13.3	12.5
		MCDBR	20.0	18.6	19.4	20.2	21.6	20.3	17.9	17.2	17.4	16.5	16.3	17.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.297	0.306	0.331	0.372	0.413	0.459	0.482	0.440	0.374	0.338	0.311	0.297
	taud		2.496	2.479	2.463	2.379	2.297	2.205	2.150	2.254	2.417	2.530	2.564	2.559
	Ebn at Noon		272	287	292	285	274	261	253	260	273	272	266	264
	Edh at Noon		23	28	31	37	41	45	47	41	33	26	22	20
(44) All-Sky Solar Radiation	RadAvg		589	866	1184	1523	1683	1837	1902	1709	1388	931	639	491
	RadStd		43	80	84	163	173	182	104	88	101	84	47	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+81

Nomenclature: See separate page